# CELLULAR SYSTEM ARCHITECTURE

# CELL

**Cell** is a basic geographic unit of a cellular system.

It comprises of small area which has a low power antenna installed in the form of a unit called the **base station**.

The location of base station depends on whether the antenna is omnidirectional or directional.

- **Center excited cells:** Uses omnidirectional antennas and the base station is at the center of the cell
- **Edge excited cells:** Uses directional antennas and the base station is at the periphery of the cell

# CLUSTER

- Each cell is assign a frequency channel or group of channels.
- Adjacent cells are assigned non – overlapping frequency channels.
- Group of such cells with non- over lapping frequency is called a cluster.
- Size of the cluster (number of cells) can be 3, 4, 7, 9, 12, 21.

The size of the cluster C, is according to shift parameters i and j.

- i is step along one direction and j is step along another direction. C

C is given as:

- $C = i^2 + ij + j^2$

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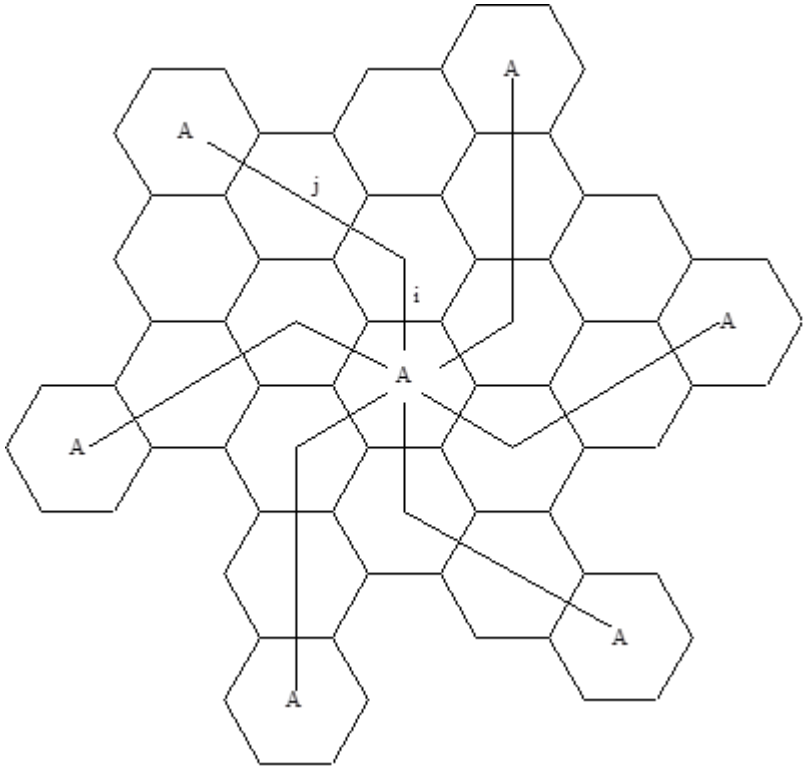
# EX

Calculate the cluster size when i=1 j=2

# CLUSTER SIZE

| Cluster Size(C) | Values of i and j |
|---|---|
| C=1 | i=1, j=0 |
| C=3 | i=1, j=1 |
| C=4 | i=2, j=0 |
| C=7 | i=2, j=1 |
| C=9 | i=3, j=0 |
| C=12 | i=0, j=2 |

# CLUSTER SIZE

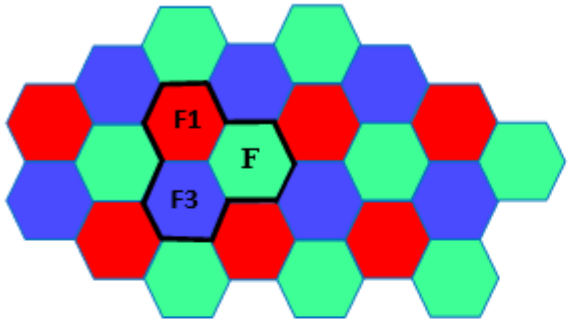


# COMMON CLUSTER SIZES

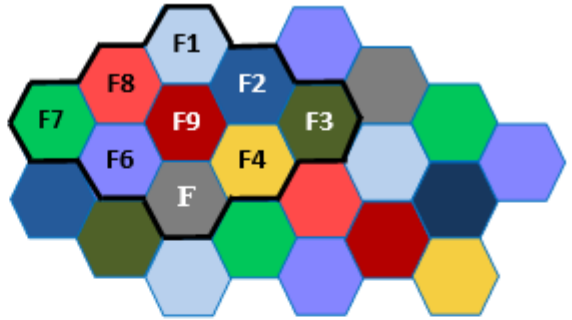
GSM cluster size C = 3 and 4

CDMA cluster size C = 1

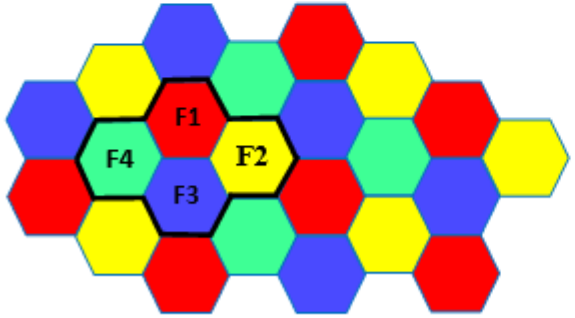
FM cluster size C = 7 or 9



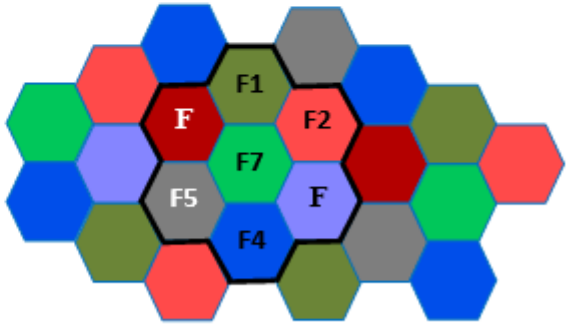

N = 3




N = 9




N = 4

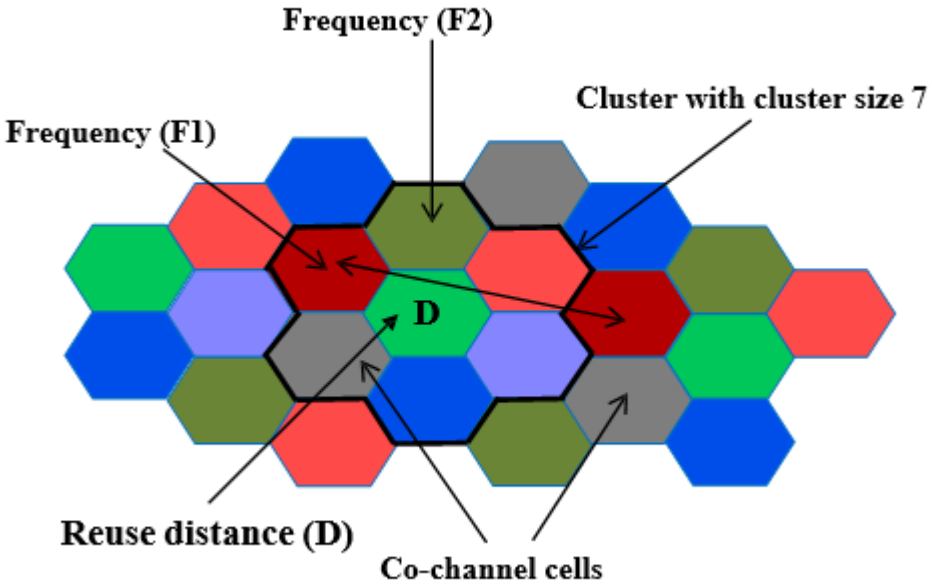



N = 7

# REUSE DISTANCE

- Same frequency (F1 and F1) can be assign to different cells in such a way that they are at a sufficient distance from each other and communication in this cell do not interfere with each other.
- Such cells are called **Co-channel cells**. If number of cells in cluster is C then frequency reuse factor is **1/C.**
- Distance between co-channel cells is called **reuse distance.**
  - Larger the cluster, more the reuse distance, less interference.
  - Smaller the cluster, lesser the reuse distance.

# REUSE DISTANCE



# SHAPE OF THE CELL

- Expected shape of the cell owing to the isotropic nature of the base station antenna should be a circle.
- Gaps results in loss of coverage and overlapping lead to interference of the signal.
- Therefore a polygon should be taken which can cover entire area without overlap and gaps and also it should rep geometry of circle.

**Equilateral triangle:** 17.77 % of area of circle

**Square:** 63.7 % of area of circle

**Hexagon:** 83% of area of circle

# SIZE OF CELL

The size of the cell can be controlled by controlling the power of antenna.

The size of the cell depends on tele density and topography within a particular region

# SIZE OF CELL

What are the key factors that determine the size of a cell in a cellular network, and how do they affect the network's coverage and capacity?

What challenges arise when deploying cellular networks in urban versus rural environments, and how does cell size vary between these settings?

# CELL SIZE

| Feature | 4G (LTE) | 5G |
|---|---|---|
| **Cell Density** | Larger cells (macro cells) | Dense deployment of small cells (microcells, picocells, femtocells) |
| **Frequency Bands** | Sub-3 GHz | Sub-6 GHz and mmWave (> 24 GHz) |
| **MIMO Technology** | Basic MIMO | Massive MIMO with advanced beamforming |
| **Latency** | 30–50 ms | As low as 1 ms |
| **Network Slicing** | Not available | Available, enabling specialized services for different use cases |
| **Capacity** | Moderate (focused on broadband) | Massive capacity (designed for IoT, mMTC, and eMBB) |
| **Handover and Mobility** | Efficient handover for mobile broadband | Enhanced handover with low latency, even at high speeds |

# SIZE OF CELL

According to size of cell, they are names as:

**Macro cells**: Macro cells used for remote or sparsely populated areas. Areas of the cells are 10km or more in diameter. Used in suburbs of cities.

**Micro cells**: – Micro Cells used in densely populate area with diameter around 10km.

**Pico cells**: Covers small areas like buildings, a small volley between mountains, a tunnel etc. where the coverage from large cells is not possible. Diameter is of few meters

**Selective cells**: Selective cells used when 360 degree coverage is not requested. They are used to fill the gapes/holes in the coverage of a BS.

# ADVANTAGE OF USING SMALL CELLS

**Higher capacity**: Huge cells allow limited number of users/km2. Small cells can allow more users with the help of SDM, frequency reuse, FDM and frequency hopping.

**Low power Tx/Rx**: Power is a critical resource for mobile stations. Base stations can easily transmit high power so that the coverage area is increase but it has ill effects on the people surrounding it. Secondly mobile devices cannot do the same. Therefore small cells which requires only low transmit power are useful.

**Reduced Interference**: Small cells means less distance more the distance covered by antenna, more is the interferenc Small cells covering small distance offer less interference.

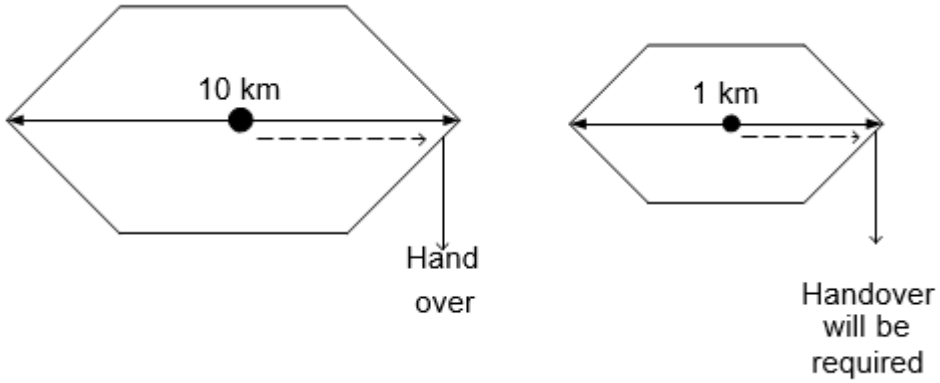
**Robustness**: If a base station covering large area fails, it influences connection of many people where as if it is divided into small cells, failure of BS of small cell will effect communication within a small area Small cells offer a decentralized system.

**Umbrella cel**ls: A heavily used road crosses an area when there are lots of microc This would required lots of handovers as people cross the road. This is handled by umbrella cells which controls all these cells

# DISADVANTAGES OF USING SMALL CELLS

**Large Infrastructure:** Smaller the cells, more the cells more BS are required and a complex infrastructure to connect these base stations are also

**Frequent Handover:** To provide uninterrupted services in form of cells or data, proper handovers are required. The frequent of handover would be more when he cell size is small and speed of movement is also more.




**Frequency Planning:** Smaller the cells, distribution of frequencies have to be done carefully to avoid the interference at the same time keeping in mind scarcity of frequencies.

# HANDOVER

When the user moves from one cell to another , while a call is in progress the channel from new cell is assigned to ensure that there is no call drop.

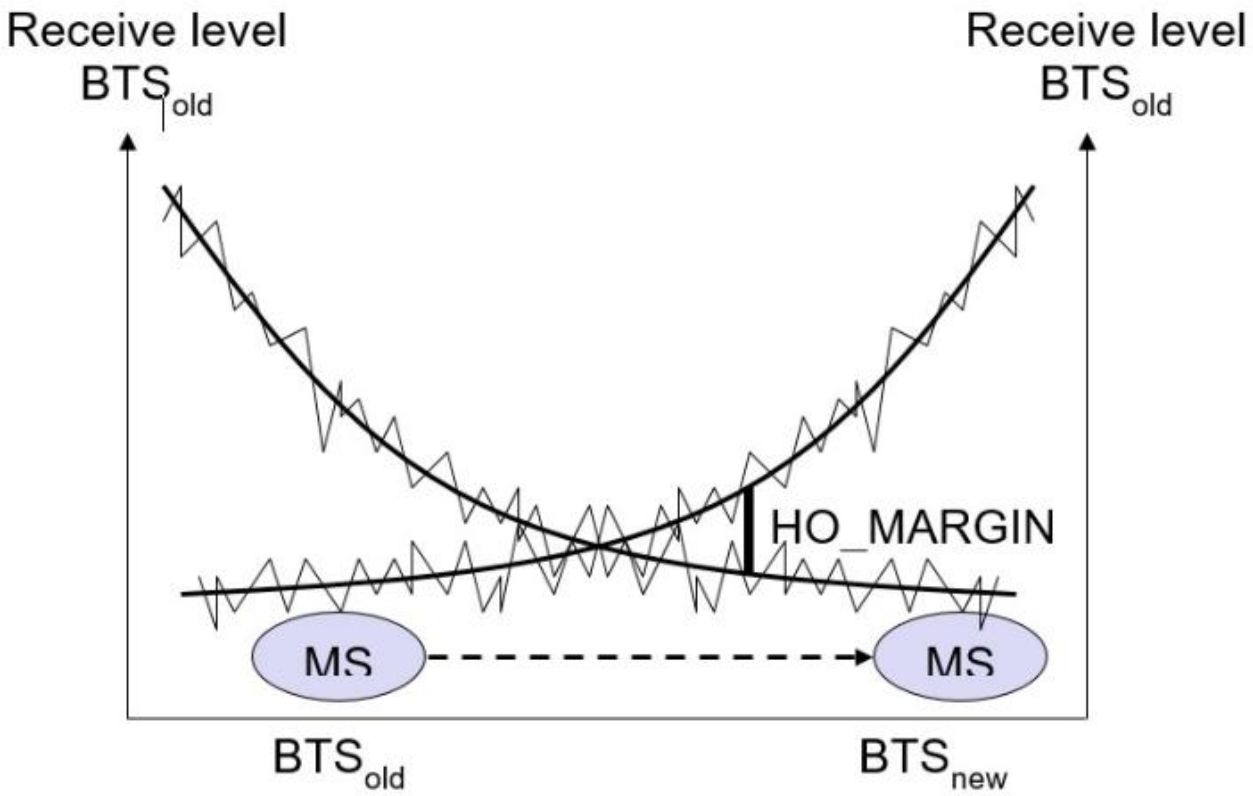
The process of handing over the call from old base station to new base station is known as handover.

For handover decision, the signal strength of the current base station, is compared with the signal strengths of the surrounding stations.

Both mobile station and base station perform quality measurement on current link as well as in the neighboring cells.

If difference of quality measurement is greater than a threshold known as HO_MARGIN and the channels are available with the new cell, handover is initiated.

# HANDOVER


Receive level
BTS old
Receive level
BTS old
HO_MARGIN
MS
MS
BTS old
BTS new